

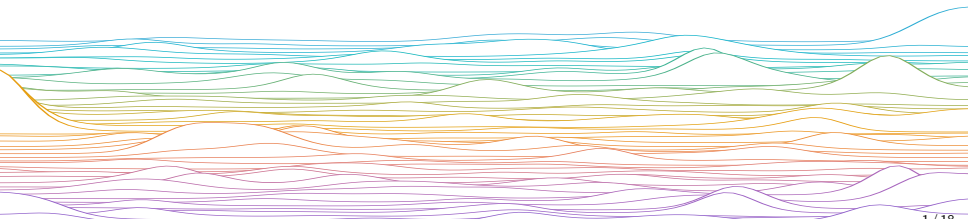
Hello and Introduction

Computational Neuroscience
University of Bristol

M Rule

Learning outcomes:

- ▶ Know where to find course material on Github
- ▶ Know that exam material will be drawn from lecture slides, homeworks, and coursenotes
- ▶ Schedule and Contacts
- ▶ Broader course context



Course materials on Github

[github.com/ engmaths/SEMT30003_2025](https://github.com/engmaths/SEMT30003_2025)

- ▶ Slides/labs/coursework/everything

Textbooks and Reading

Material will be accompanied by reading or exercises taken from

- ▶ *Conor Houghton's Coursenotes (on course github)*
- ▶ *Neuronal Dynamics* (Gerstner, Kistler, Naud, Paninski), [free online](#).
- ▶ *Theoretical Neuroscience* (Abbott, Dayan)

Exam material will be drawn from lecture slides, homeworks (i.e. labs, formative coursework), and coursenotes

Unit variants

CS minor:

- ▶ *Topics in Computer Science* exam question (lectures & labs, term weeks 1–8).

SEMT:

- ▶ 50% Final (lectures & labs term weeks 1–8)
- ▶ 50% extended coursework (term weeks 9–11)

Staff Contacts (via Teams/Email)

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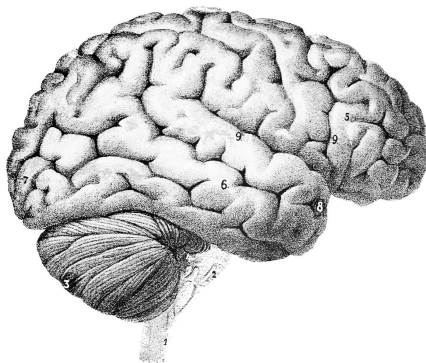
Why Brains?

1–2 kg ($\approx$ 2% body weight)

20% of caloric expenditure

Azevedo et al. 2009¹:

- ▶ $\sim 86 \cdot 10^9$ neurons
in human brain



*Outer surface of the human brain, Sanger Brown, c. 1894,
Wikimedia*

¹Azevedo, Frederico AC, et al. "Equal numbers of neuronal and non-neuronal cells make the human brain an isometrically scaled-up primate brain." *Journal of Comparative Neurology* 513.5 (2009): 532-541.

Why Brains?

La fixité du milieu [intérieur] suppose un perfectionnement de l'organisme tel que les variations externes soient à chaque instant compensées et équilibrées.

Bien loin, par conséquent, que l'animal élevé soit indifférent au monde extérieur, il est au contraire dans une étroite et savante relation avec lui, ...

— Claude Bernard (1878) *Leçons sur les phénomènes de la vie communs aux animaux et aux végétaux, Tome I*

Paraphrasing:

- ▶ The stability of an organism's internal environment requires physiological processes that compensate for external disturbances.
- ▶ Organisms with higher cognitive capacities are not more removed from the external world, but rather more attuned to it.

⋮

< 1900 Mathematical Physiology

Harvey (1628) Fick (1855) Frank (1895) Starling (1896)
1878: Claude Bernard, *Milieu Intérieur*

1932 Homeostasis

Walter Cannon

1924 Quantitative Experiments

Hans Berger, Electroencephalography (EEG)

1952 Mathematical Neurophysiology

Hodgkin & Huxley

1940–60 Cybernetics

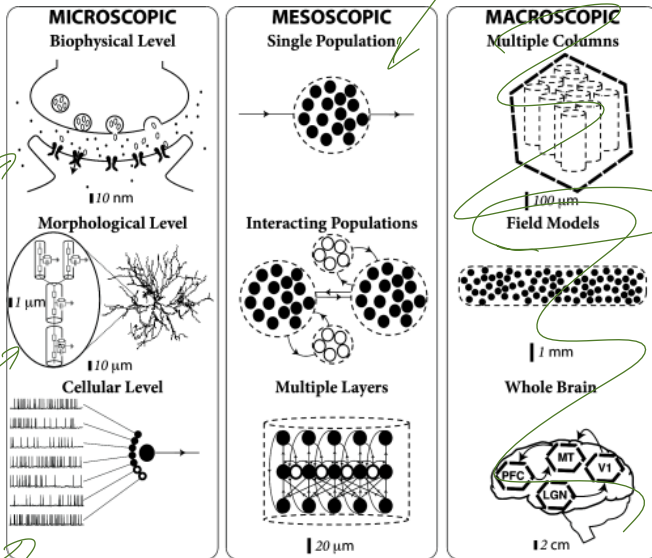
Wiener ('48), McCulloch & Pitts ('43), Rosenblatt ('58), Turing, von Neumann, Minsky

⋮

There are now in the world machines that think, that learn and that create. Moreover, their ability to do these things is going to increase rapidly until—in a visible future—the range of problems they can handle will be coextensive with the range to which the human mind has been applied.

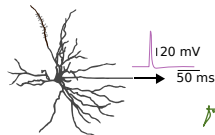
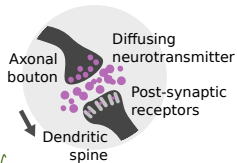
— Herbert A. Simon, Allen Newell (1958)¹

¹ Simon, H.A. and Newell, A. (1958) "Heuristic problem solving". Operations Research 6(1), pp.1-10.

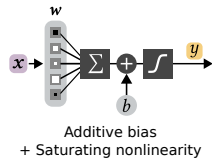
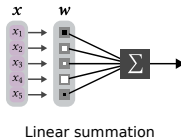
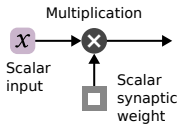


Gerstner & al. Neural Dynamics Fig 12.1

Biological



Artificial



Modelling Scales

... Quantum Chemistry, Molecular dynamics

Molecules
Gillespie, Master Equation

Concentrations
Mass-Action Kinetics

Conductance Models
Hodgkin-Huxley

Spiking Models
Leaky Integrate and Fire

Rate Neurons
Neural Mass/Field Models

Poisson Neurons
Generalized Linear Models

Binary Neurons
McCulloch-Pitts, Hopfield, Perceptron

Cognitive Neuroscience, Psychology ...

Physiological,
Quantitative

Biological
Realism,
Data needed
to identify
parameters



Computational
Efficiency,
Mathematical
Tractability



Phenomenological,
Qualitative

data analysis

Scheduled Topics

Week

- 1** Intro to neuroscience & background
- 2** The action potential
- 3** Synaptic communication
- 4** Plasticity 1: Cerebellum & perceptron
- 5** Plasticity 2: Hippocampus & hopfield Networks
(reading week break)
- 6** Rate models and vision
- 7** Statistical models and data analysis
- 8 (extended coursework — to be submitted 50% mark)
- 9 (extended coursework ...)
- 10 (extended coursework ...)

end

